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# Rogar induced changes in the free amino acid pool of haemolymph of diving beetle, *Dytiscus verticalis Say* (Coleoptera : Dytiscidae)

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**Abstract :** In the haemolymph of control *Dytiscus verticalis,* twenty free amino acids were detected by two dimensional paper chromatography and their concentrations were determined by spectrophotometer. Glutamine occurred in the highest concentration followed by glycine. The concentration of all free amino acids except that of ornithine decreased after one hour of the rogar treatment. After one hour and forty five minutes, concentration of majority of amino acids decreased except for the glutamic acid, leucine and lysine, which showed an increasing trend after one hour of treatment.

**Keywords: Rogar, amino acid, haemolymph, *Dytiscus*.**

## INTRODUCTION

Amino acids are the building blocks of proteins which contribute in the formation of structure of living system. The high concentration of free amino acids is believed to play an important role in osmoregulation (Bishop et.al, 1926 and Beadle and Shaw, 1980)[1,2]. Insect's structure and functions are greatly governed by such important biomolecules and any type of chemical as well as insecticidal exposure may create tremendous changes in their organisation[3].

The present study is aimed to detect the rogar induced changes in the qualitative and quantitative concentration of the amino acids forming the "pool" in the haemolymph of *Dytiscus verticalis* Say at fixed time intervals of rogar treatment and there by also to observe the corelated changes in their behaviour.

The experimental inset is a coleopteran beetle, leading a vertical diving life style in order to predate upon the small column floating diatoms, zooplanktons & small larvae of fishes. It is also behaviourly supposed to be fresh water scavenger.

Rogar is an agriculture pesticide commonly used in paddy, maize and other khariff crop fields adjoining the fresh water bodies or ponds there by slowly travelling into the water and damaging the flora & founa including the beetles.

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## MATERIALS AND METHODS

*Dytiscus verticalis* adults were collected from local ponds, washed with distilled water to remove all adhering debris. The haemolymph was collected by cutting the coxae from the body wall. According to the method of Pant and Agrawal (1964)[3], 1 ml. of haemolymph was pooled and deprotenized. All the chemicals used were of the analytical reagent grade. Extracts were applied as compact spots on whatman No. 1 filter paper sheet (40 cm x 40 cm.); the two dimensional paper chromatography was done according to the method of smith and Agiza (1951)[4]; and the chromatograms were first developed for six hours with n-butanol, glacial acetic acid, and distilled water (4:1:5). The second solvent was made with phenol saturated with distilled water, spots were developed with nirhydrin and identified with predetermined values of the known amino acids.

The concentration of all free amino acids were determined by elution. The ninhydrin coloured spots were placed separately in tubes, to which were added one ml NaoH (neutralized with phenolphthalene), one ml. citrate buffer (pH-5) and two ml. ninhydrin solution. The tubes

were then placed in boiling water , to which was added one ml stannouschloride solution, this produced red colour after fifteen minutes; their content was made up to volume of 10 ml. with saturated solution of NaCl and 5 ml. of n-butanol was added to each tube, which was then shaken and kept ready for estimation. The optical density of colour was measured with a carl zeiss universal spectrophotometer (Model VSU-1) at 570μ and at 440 μ (only for proline); against a blank extract prepared from a piece of the same filter paper of the same areas. The concentrations of the separated amino acids were calculated from standard curves; previously prepared by using standard amino acids solutions chromatographed for the same period using the same solvent.

The $LC_{50}$ value of rogar for the insects under investigation was determined by the following method. A solution of 50 percent malathion was prepared and 2 ml of this 50 percent malathion was added to 100ml of acetone to make 1 percent solution. From this 0.1%, 0.05%, 0.04%, 0.02%, and 0.01% solution of malathion were prepared and stocked to find out $LC_{50}$ value.

*Dytiscus verticalis* is an aquatic insect. Ten beakers, each of which contained 10 cc. of the pond water with one individual, was brought to the laboratory. One ml. of rogar solution was injected in to the abdomen of these insects - one concentration in each of the ten separately, the insects were then dried-one concentration in each of the ten separately. The insects were then dried under an electric fan for a few minutes (Sarup et. al ; 1971) and transferred again to water. Mortality count was made after twenty four hours to determine the toxic effect of each concentration, the moribund insects also being counted as dead. The treatment with each concentration was repeated thrice. For control, only acetone was injected in to the abdomen of some insects and the data obtained by probit analysis (Finney 1952).

Known quantity ($LC_{50}$) of rogar was sprayed on the insects and haemolymph was pooled after the lapse of one hour of insecticidal treatment by cutting the coxae. Again after the Lapse of one hour and forty five minutes the haemolymph was collected for chromatographic separation and quantitative estimation was carried out by processes describe above.

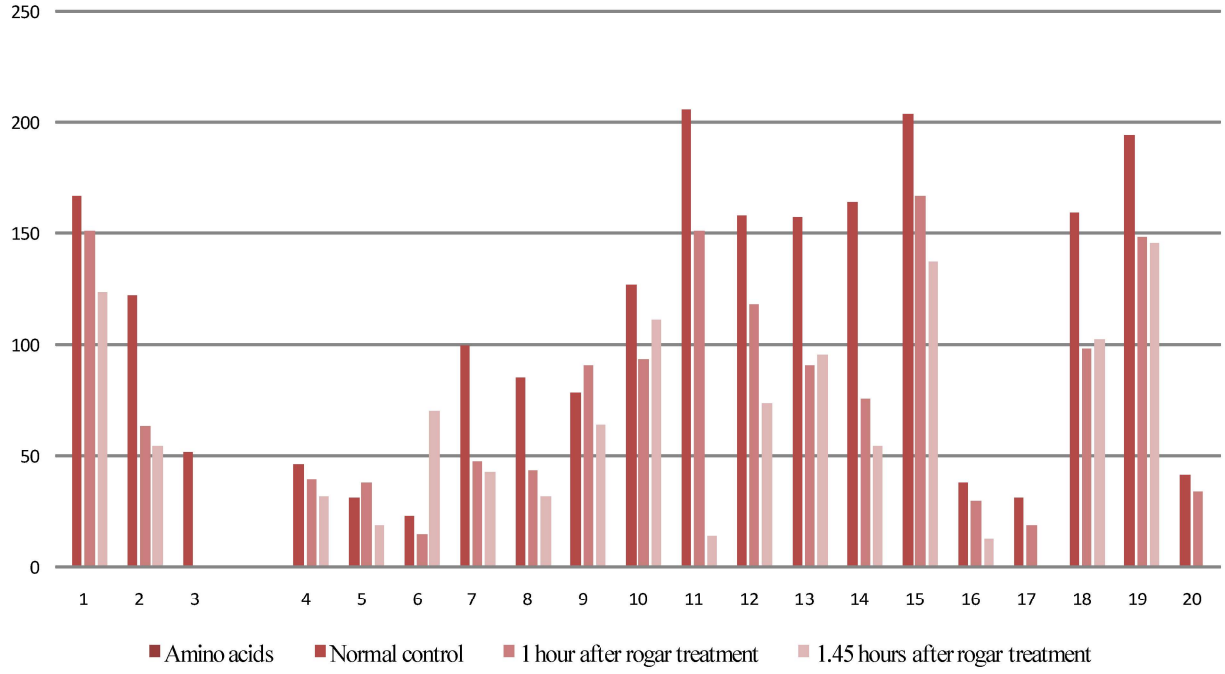
**Table -1: Comparative concentration of free amino acids in the haemolymph of *Dytiscus verticalis* in Control and treated specimens.**

**(concentration : μ mol /100ml of haemolymph)**

| | Amino acids | Normal control | 1 hour after rogar treatment | 1.45 hours after rogar treatment |
|---|---|---|---|---|
| 1. | Alanine | 166.88 | 151.47 | 123.58 |
| 2. | Arginine | 121.67 | 62.88 | 53.92 |
| 3. | α-amino-n butyric acid | 51.26 | –* | –* |
| 4. | v-amino-n butyric acid | 45.66 | 39.17 | 31.22 |
| 5. | Aspartic acid | 30.83 | 37.88 | 18.80 |
| 6. | Cystine | 22.51 | 14.11 | 69.53 |
| 7. | Histidin | 99.19 | 47.58 | 42.32 |
| 8. | Hydroxylysine | 84.83 | 42.88 | 31.60 |
| 9. | Ornithine | 77.88 | 90.38 | 63.47 |
| 10. | Lysine | 126.88 | 93.27 | 111.22 |
| 11. | Glutamine | 206.40 | 151.38 | 13.57 |
| 12. | Valine | 158.27 | 117.94 | 73.00 |
| 13. | Leucine | 157.55 | 90.67 | 95.12 |
| 14. | Proline | 164.52 | 75.13 | 53.92 |
| 15. | Glycine | 204.40 | 167.33 | 136.93 |
| 16. | Phenylalanine | 37.40 | 29.23 | 12.41 |
| 17. | Tyrosine | 30.88 | 18.27 | –* |
| 18. | Glutamic acid | 159.46 | 97.95 | 101.91 |
| 19. | Serine | 194.66 | 148.38 | 145.90 |
| 20. | β-alanine | 40.76 | 33.32 | –* |

**–* Denotes the absence of amino acid**

**Fig.-1 : Histogram showing rogar induced variation in the amino acids of haemolymph of D. *verticalis* with reference to the control peak.**




## OBSERVATIONS

In *Dytiscus verticalis* twenty free amino acids were identified by two-dimensionl paper chromatogram, their relative concentrations are shown in Table-1. Glutamine showed the highest concentration and next to that was Glycine, serine and alanine also occurred in high concentrations. Only one sulphur containing amino acid, cystine, was found in low concentration.

After one hour forty five minutes of rogar treatment the concentration of all the amino acids except glumatic acid, lysine and leucine showed a decrease.

## DISCUSSION

All the amino acids commonly found in proteins have been identified in the insects haemolymph, although some of them such as cystine, serine, hydroxyproline and tryptophane are found only in few insects or in low concentrations (Gilmour, 1965).[5]

In *Dytiscus verticalis* glutamine shows a high concentration. Leven book (1962) reported that glutamine relative concentrations in insect tissue are considerably more variable , the ratio of glutamine over glutamate in locust, haemolymph is 9 : 1 (Kilby, 1958).

Cystine, although in low concentration, do occur in *D. verticalis*. According to Gilmour (1961), cystine is interconvertible by oxidation and because cystine is easily oxidised aerobically to cystine and cysteic acid, the isolation of these last two is not a proof of their presence in the living organism (Robinson, 1963).

Belogun (1974) has suggested that proline plays the role of a readily available metabolized energy reserve for flight and movements. Besides, a great number of amino acids like alanine, glutamine, glycine and proline are known to have an important role in the synthesis of cuticle proteins, chitin and other constituents of cuticle.

In the present investigation malathion poisoned insects showed depletion of all the amino acids except ornithine. This suggests that the levels of the free amino acids in the intoxicated insects are influenced by the amount of the insecticides used-the sub- lethal or lethaldose and the time at which toxic effects are evaluated. Winteringham and Harrison (1956) reported depletioa of glutamate and proline in DPP poisoned house-flies, the liberated ammonia being trapped by glutamine. Corrigan and Kearns (1963)[7] observed that the concentration of glutamine and proline in the DDT poisoned cockroaches varies inversely with the increase in intoxication. In the present study also the concentration of proline and glutamine decreased with the increase in intoxication.

Man Singh (1964)[8] has described the free

amino acids in the haemolymph of the malathion resistant *Blatella germanica*. According to him, there is no change in the concentration of amino acids in the haemolymph of insects. In the present investigation, however the insects treated with rogar definitely showed a marked change in the concentration of the free amino acids in the haemolymph. The depletion of amino acids in probably a consequence of higher metabolic activity in the poisoned insects which causes an imbalance between the rates of anabolism and catabolism in them.

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